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import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score


dataset = pd.read_csv("Iris.csv")

print("First 5 rows:\n", dataset.head())


# Drop non-numeric columns (Id and Species)
X = dataset.iloc[:, 1:5].values   # Sepal & Petal features

# Feature Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)


wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)


plt.plot(range(1, 11), wcss)
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()


# From elbow graph → optimal clusters = 3

kmeans = KMeans(n_clusters=3, init='k-means++', random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)


score = silhouette_score(X_scaled, y_kmeans)
print("Silhouette Score:", score)

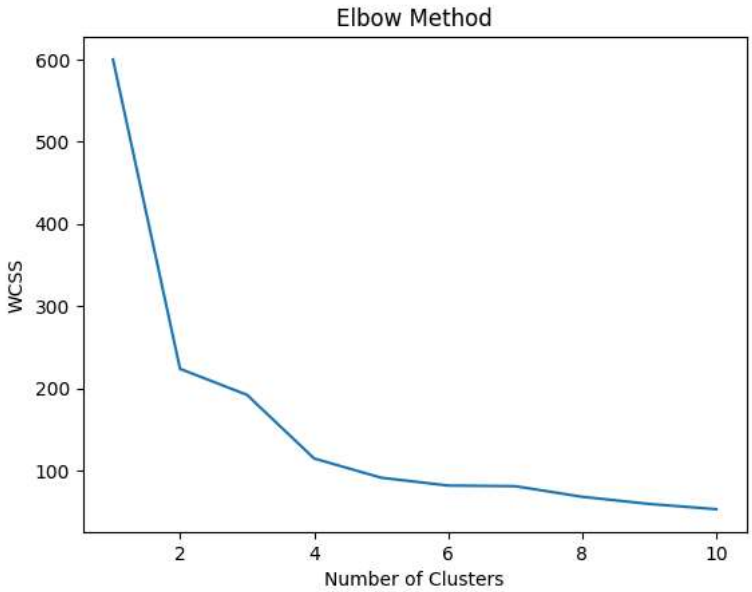


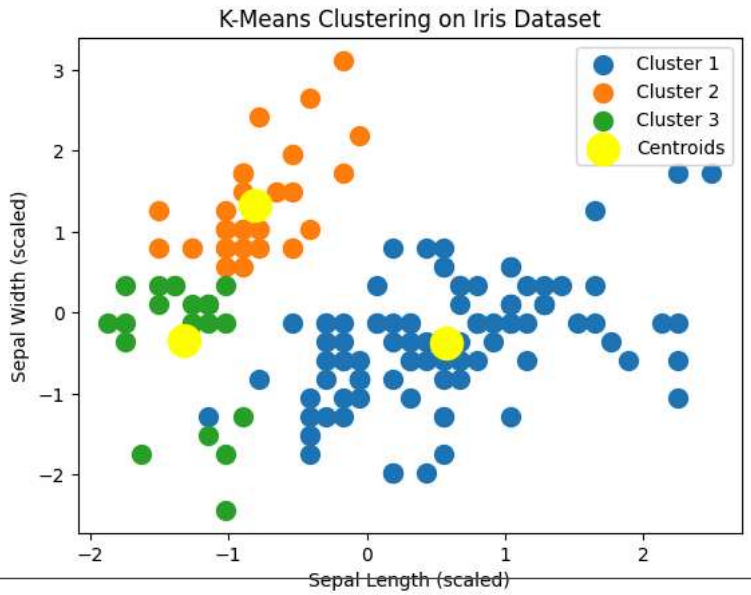
plt.scatter(X_scaled[y_kmeans == 0, 0], X_scaled[y_kmeans == 0, 1], s=100, label='Cluster 1')
plt.scatter(X_scaled[y_kmeans == 1, 0], X_scaled[y_kmeans == 1, 1], s=100, label='Cluster 2')
plt.scatter(X_scaled[y_kmeans == 2, 0], X_scaled[y_kmeans == 2, 1], s=100, label='Cluster 3')

# Centroids
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1],
            s=300, c='yellow', label='Centroids')

plt.title("K-Means Clustering on Iris Dataset")
plt.xlabel("Sepal Length (scaled)")
plt.ylabel("Sepal Width (scaled)")
plt.legend()
plt.show()
```

```
First 5 rows:
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa
```




```
Silhouette Score: 0.4787241921049546
```